Filter-matched rate calibration for variable-rate PING training

exp077 · 5 August 2026 · Draft

Abstract

This staged experiment tests which Poisson rates preserve MNIST class evidence. An artificial neural network (ANN) is planned to classify static features formed by passing each pixel's spike train through an AMPA synapse and non-spiking membrane, then averaging its voltage over the presentation.

ANN accuracy against rate remains the planned primary result; a linear transfer-function calculation would check the empirical feature variance. No rate recommendation was reached because the response-library pilot failed.

Step 1 is complete. The local generator passed all 8 focused checks, including agreement with the shared `tools/snn` cell below the registered numerical tolerance. Step 2 then stopped at its locked convergence gate: none of the candidate draw counts through K = 512 stabilized both empirical moments. No final response library or later stage was run.

Purpose and scope

The question is:

> At a given encoding rate, does the temporally filtered image still contain enough information for an ANN to recognize the digit?

The target PING architecture has 784 pixel channels feeding conductance-based excitatory cells. Its fixed synaptic and membrane equations define the features, without trained input, recurrent, or output weights. The ANN thus measures filtered-input decodability independently of a trained PING network.

The experimental logic is:

1. Convert 0.25–25 Hz Poisson inputs into static, filter-matched features.
2. Measure their distributions and test a linear variance prediction.
3. Train the ANN across the rate grid with the held-out test set sealed.
4. Measure ANN accuracy and locate its lowest decodable rates.

- Nonlinear ANN above chance: this decoder can access class evidence.
- Nonlinear ANN at chance: this representation and decoder cannot extract reliable evidence.
- Nonlinear ANN succeeds but linear decoder fails: the evidence is not linearly accessible.

The ANN threshold is a decoder-relative decodability edge, not an absolute information boundary or a prediction of PING accuracy. The practical training floor instead applies a predeclared useful-accuracy criterion. A later experiment must test whether PING can train across the selected range.

Methods

1. **Generated and validated filter-matched pixel features.** Every image used a 200 ms presentation and 0.1 ms timestep. The tested rates are 0.25, 0.5, 0.75, 1, 1.5, 2, 2.5, 3, 4, 5, 10, 25 Hz; dense sampling below 5 Hz resolves the lower edge, while 25 Hz is the proposed later training ceiling. Training samples uniformly from these points. Seeds 42, 43, 44 define independent feature, response-library, and ANN runs.

   `protocol.json` records the grid and deterministic MNIST train, validation, and held-out indices. Pixel-intensities x remain in [0, 1]. If r is the encoding rate at pixel-intensity one, the expected pixel rate is r x.

   Each pixel drives an independent AMPA synapse and uncoupled, non-spiking excitatory membrane, capturing AMPA decay, conductance-dependent integration, and within-window timing that raw counts omit. This probe has no threshold, reset, recurrence, or trained weights. It uses the target PING cell's decay-then-add synapse and exponential-Euler membrane update, implemented locally without changing the shared simulator.

   The complete feature path is

   $$S_i(t) \to g_i^{\text{pix}}(t) \to v_i(t) \to z_i \to \text{ANN}. \quad (1)$$

   Here S_i, g^{pix}_i, v_i, and z_i are pixel i's spike train, AMPA conductance, subthreshold voltage, and time-averaged feature; t is time and ANN is the classifier.

   At each timestep, we drew the encoded input according to

   $$S_i(t) \sim \text{Bernoulli}(r\Delta t x_i). \quad (2)$$

   Here x_i is grayscale pixel-intensity, r is the encoding rate in spikes per second at pixel-intensity one, Δt is the timestep in seconds, and Bernoulli is an independent binary draw. Configurations with probability above one are invalid.

   We updated the AMPA conductance using

   $$g_i^{\text{pix}}(t) = \beta_{\text{AMPA}} g_i^{\text{pix}}(t-1) + w_{\text{probe}} S_i(t). \quad (3)$$

   $$\beta_{\text{AMPA}} = \exp\left(-\frac{\Delta t}{\tau_{\text{AMPA}}}\right). \quad (4)$$

   Here β_{AMPA} is one-timestep conductance retention, τ_{AMPA} is AMPA decay time, w_{probe} is conductance added per spike, and exp is the exponential function. Other symbols follow Equation 2. The primary 1.2 μS probe is the target architecture's nominal mean initial pixel-to-excitatory conductance; 0.6 and 2.4 μS test half and double scale.

   We integrated the membrane without thresholding or resetting:

   $$C_E \frac{dv_i}{dt} = g_{\text{L,E}}(E_L - v_i) + g_i^{\text{pix}}(t)(E_e - v_i). \quad (5)$$

$$\tau_{\text{eff}}, i(t) = \frac{C_E}{g_{\text{L,E}} + g_i^{\text{pix}}(t)}. \quad (6)$$

Here C_E is excitatory capacitance; $g_{L,E}$ is leak conductance; E_L and E_e are leak and AMPA reversal potentials; and $\tau_{eff,i}$ is the instantaneous conductance-dependent time constant. Other symbols follow Equation 1.

We initialized voltage at E_L and conductance at zero, simulated one complete presentation, and calculated

$$z_i = \frac{1}{T}\int_0^T \left(v_i(t) - E_L\right) \mathrm{d}t. \quad (7)$$

Here T is presentation duration and z_i is mean baseline-subtracted voltage. We did not divide by rate or disclose it to either decoder, so lower rates retain their weaker, noisier signal.

Focused tests covered encoder probability, count moments, pixel independence, AMPA decay, exponential-Euler voltage updates, deterministic replay, agreement with one uncoupled target-PING excitatory cell, and spike-timing sensitivity. All 8 checks passed. At the seeded count validation point, the expected and observed mean counts were 4 and 4.01 the largest absolute inter-channel count correlation was 0.023.

The corresponding plot is defined in Results, Step 1.

2. **Attempted the empirical response-library convergence gate.** Before generating a final library, we locked candidate draw counts K of 64, 128, 256, 512 and a hard maximum of 512. The deterministic pilot covered 270 conditions: six intensities, five rates, all three probe conductances, and all three registered seeds. For each candidate, we compared the first K draws with the next K independent draws.

   Mean and unbiased-variance discrepancies were divided by locked absolute or relative tolerances, whichever was larger. Both metrics had to keep their 95th percentile at or below 1 and their maximum at or below 2. The smallest passing K would have selected the final draw count. If none passed at the maximum, the protocol required stopping without changing the rule.

   The pilot reached that stop. We therefore did not run K independent Step 1 draws for the complete set of 256 grayscale levels, 12 rates, and three probe conductances.

   The planned final library would estimate the conditional mean

$$\hat{\mu}_z\left(x, r, w_{\text{probe}}\right) = \frac{1}{K}\sum_{k=1}^{K} z^k. \quad (8)$$

   and conditional variance

$$\hat{\sigma}_z^2\left(x, r, w_{\text{probe}}\right) = \frac{1}{K-1}\sum_{k=1}^{K}\left(z^k - \hat{\mu}_z\right)^2. \quad (9)$$

   Here x, r, and w_{probe} are pixel-intensity, rate, and probe conductance; $z^{(k)}$ is draw k; K is the draw count; and Equations 8 and 9 estimate the conditional mean and variance.

Had the pilot passed, we would have retained all K simulated values as the primary empirical response library. During later ANN feature generation, the registered design would draw one value according to

$$J \sim \text{DiscreteUniform}(1, \cdots, K), \quad z_{\text{sample}} = z^{J}. \quad (10)$$

Here J is uniform on the K draws and z_{sample} is the sampled feature; other symbols follow Equations 8 and 9. Mean and variance summarize noise, but the ANN samples the retained empirical values because low-rate responses may be zero-heavy, skewed, and non-Gaussian.

Because the gate failed, the committed artifacts contain the locked pilot, its complete convergence trajectory, seed recipe, and regeneration command, but no response-library manifest or array. The ungenerated calibration has at most 9216 conditions.

The corresponding plot is defined in Results, Step 2.

3. **Compare the empirical variance with a linear-filter prediction.** This supplementary check asks whether a local linear approximation explains the Step 2 feature variance. It will not set the training range.

   For mean spike rate λ, stationary mean conductance is

$$|g|_{\lambda} = \lambda w_{\text{probe}} \tau_{\text{AMPA}}. \quad (11)$$

   and stationary mean voltage is

$$|v|_{\lambda} = \frac{g_{\text{L,E}} E_L + |g|_{\lambda} E_e}{g_{\text{L,E}} + |g|_{\lambda}}. \quad (12)$$

   Here λ is encoding rate times pixel-intensity; bar(g)$_{\lambda}$ and bar(v)$_{\lambda}$ are stationary mean conductance and voltage. Other symbols follow Step 1.

   Linearizing the synapse and membrane around that operating point gives

$$G_{\lambda(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \cdot \frac{E_e - |v|_{\lambda}}{i\omega C_E + g_{\text{L,E}} + |g|_{\lambda}}. \quad (13)$$

   Here G_{λ} (ω) is the local synapse-plus-membrane transfer function; ω is angular frequency; and i is the imaginary unit. Because mean conductance changes gain and effective membrane time constant, Equation 13 defines a family of responses.

   The finite averaging window contributes

$$A_{T(\omega)} = \frac{1 - \exp(-i\omega T)}{i\omega T}. \quad (14)$$

$$H_{\lambda(\omega)} = A_{T(\omega)} G_{\lambda(\omega)}. \quad (15)$$

   Here A_T (ω) is the duration-T averaging response; H_{λ} (ω) is the complete response from input fluctuation to averaged feature; T is presentation duration; and exp is the exponential function.

   For centred ideal Poisson input,

$$S_{\mathrm{in}(\omega)} = \lambda. \quad (16)$$

$$S_{z(\omega)} = \left|H_{\lambda(\omega)}\right|^2 S_{\mathrm{in}(\omega)}. \quad (17)$$

$$\mathrm{Var}_{\mathrm{linear}(z)} = \frac{1}{2\pi}\int_{-\infty}^{\infty} \left|H_{\lambda(\omega)}\right|^2 S_{\mathrm{in}(\omega)}\,\mathrm{d}\omega. \quad (18)$$

Here S_{in} (ω) and S_z (ω) are input and predicted output power spectral densities; $|H_\lambda (\omega)|^2$ is transmitted noise power; Var_{linear} (z) is predicted feature variance; and π is the circle constant. Appendix A derives Equations 11–18.

We will evaluate every distinct calibration point, compare predicted with empirical Step 2 variance, and validate low-, middle-, and high-drive gains by sinusoidally modulating the numerical probe. This stationary, continuous-time, linearized Poisson model approximates the finite, discrete Bernoulli simulation and remains diagnostic only.

The corresponding plot is defined in Results, Step 3.

4. **Construct and validate complete feature images.** For each MNIST image, we will select one of the 12 rates uniformly and sample each pixel from Equation 10, resampling every training epoch. Sampling is independent across pixels because each channel and probe is independent.

   Before fitting, the recorded MNIST training partition will be split into decoder-training and validation subsets; the test set remains sealed. On predeclared images at low, transitional, and high rates, the empirical sampler must match direct Step 1 simulation in pixel and image moments, zero fractions, and representative images.

   This validation must pass before ANN training. The corresponding plot is defined in Results, Step 4.

5. **Train the mixed-rate decoders.** We will train a primary ANN with 784 inputs, one 1,024-unit rectified-linear hidden layer, and ten outputs. The ANN learns both weight matrices from voltage features. A regularized linear softmax decoder on the same images tests linear accessibility. Neither model receives the rate or pretrained weights.

   Seeds 42, 43, 44 use Adam, learning rate 0.001, batch size 256, and initially at most 15 epochs. Every presentation samples a rate uniformly and regenerates features. Validation alone controls model selection, regularization, early stopping, and any epoch extension; test data remain sealed until Step 6. We will record each configuration, selected epoch, and training history.

   The primary ensemble uses the 1.2 μS probe; separate 0.6 and 2.4 μS runs test sensitivity. Conductance conditions are neither pooled nor disclosed.

   The corresponding plot is defined in Results, Step 5.

6. **Generate inference-only psychometric curves and choose two rate thresholds.** We will freeze all decoders before testing. At every tested rate, seeds use the same held-out images and fixed feature draws; additional draws measure encoding variability. The primary curve is

$$A_r(r) = P(\text{correct} \mid r, \text{mixed-rate nonlinear decoder}). \quad (19)$$

Here A_r is held-out nonlinear-ANN accuracy, P is correct-classification probability, and r is rate. The linear decoder is diagnostic only.

Bootstrap resampling of held-out images, probe draws, response-library simulations, and decoder seeds gives lower confidence bound L_r. The decoder-relative edge is

$$r_{\text{decode}} = \text{lowest} \ \ r \in \mathcal{R} \ \ \text{satisfying} \ \ L_r(r) > \frac{1}{N_{\text{class}}}. \quad (20)$$

The practical training floor is

$$r_{\text{train}} = \text{lowest} \ \ r \in \mathcal{R} \ \ \text{satisfying} \ \ L_r(r) \geq a_{\text{use}}. \quad (21)$$

Here $\mathcal{R}$ is the tested grid; N_{class} is the number of classes; r_{decode} is the lowest rate reliably above chance; a_{use} is the predeclared 50% useful-accuracy target; and r_{train} is the lowest rate whose lower bound reaches it. The target is locked before testing.

Primary thresholds use the 1.2 μS nonlinear ANN. The linear decoder and 0.6 and 2.4 μS runs show sensitivity to decoder capacity and conductance. We will not treat interpolated rates as observations.

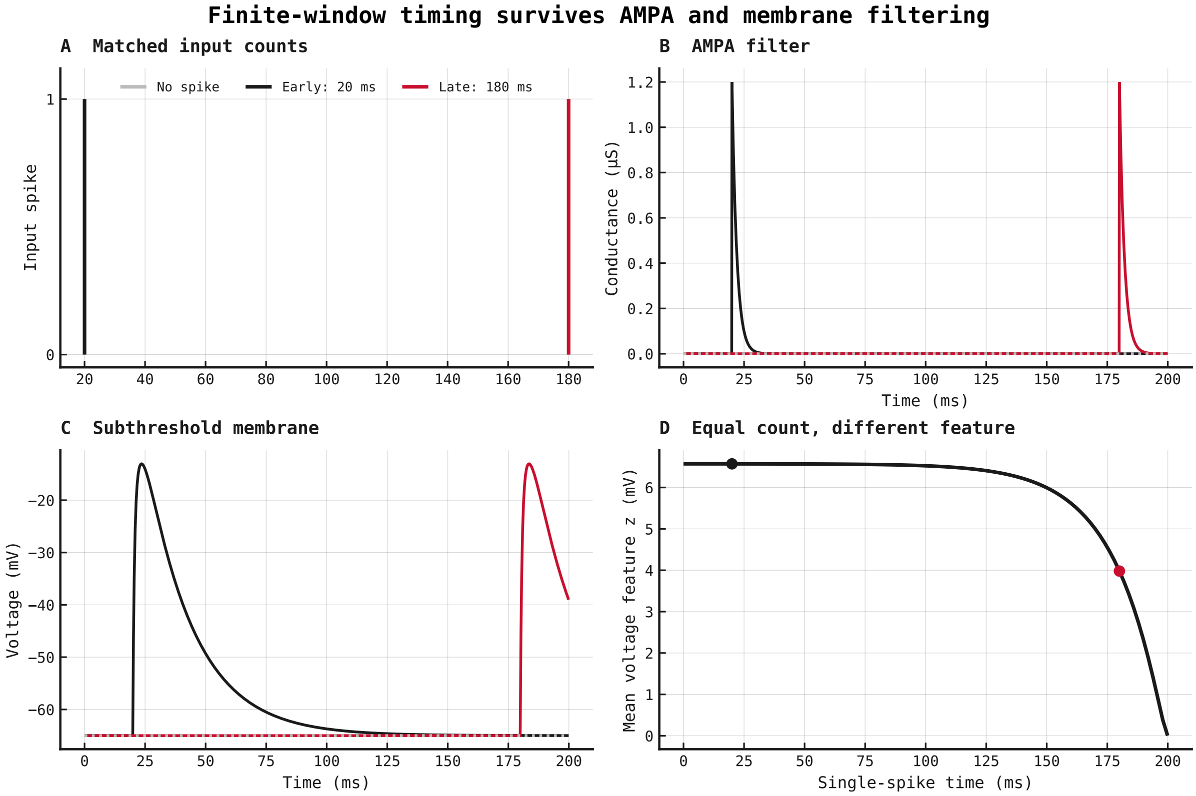
The corresponding plot is defined in Results, Step 6.

7. **Report the training-range decision and stop.** We will write `decision.json` with r_{decode}, r_{train}, their uncertainty, the probe-conductance sensitivity, decoder and artifact hashes, and all rule outcomes. The recommended later PING range is r_{train}–25 Hz. If the floor shifts by more than one adjacent grid point across conductances, we will report plausible floors rather than one value. This experiment stops at the recommendation; a separate experiment must train and test PING across it.

Results

Step 1 is complete. Step 2 records a killed convergence attempt; pending panels in later steps imply no result.

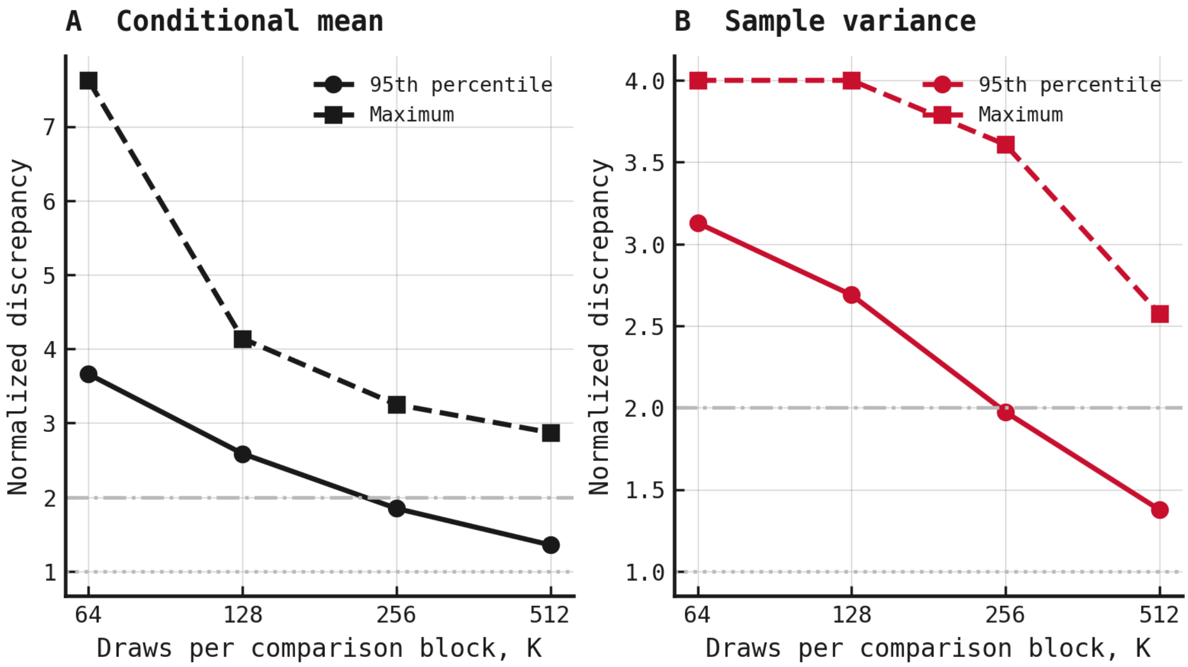
Step 1: filter-matched feature generation



Finite-window probe dynamics. Panels A–C show input spikes, AMPA conductance, and subthreshold voltage for no-spike, early-spike, and late-spike inputs. Panel D places the single spike across the presentation and reports the resulting mean voltage feature. With the same one-spike count, a spike at 20 ms produced z = 6.571 mV, whereas one at 180 ms produced 3.98 mV. Earlier input therefore contributed 2.591 mV more to the finite-window average.

The decay-then-add check reproduced the first two conductances with a maximum absolute error of 0 µS. The independently calculated exponential-Euler step and the complete local trajectory both matched the existing `tools/snn` uncoupled cell within their registered machine-precision tolerances. This validates the Step 1 feature generator only; it does not establish MNIST decodability or select a PING input-rate range.

Step 2: empirical response library



Locked response-library convergence pilot. Panel A shows normalized discrepancies between two independent K-draw estimates of the conditional mean; panel B shows the same comparison for unbiased sample variance. Solid circles report the 95th percentile and dashed squares the maximum across the non-zero-intensity evaluation conditions. The horizontal rules mark the locked limits of 1 and 2, respectively. At $K = 512$, the mean discrepancies were 1.356 at the 95th percentile and 2.87 at maximum; the variance discrepancies were 1.377 and 2.575. Neither moment passed, so no K was selected and the full empirical library was not generated.

Zero intensity returned the resting feature exactly, but this structural check could not rescue the failed convergence rule. Step 2 therefore establishes only that K through 512 was insufficient under the registered tolerances. It does not establish empirical response distributions, MNIST decodability, a training-rate floor, or PING accuracy.

Step 3: linear-filter prediction

Pending plot. Pending Step 3. Planned file: `linear_filter.svg`.
Planned caption. Linear response and variance prediction. A log-frequency (Hz) Bode panel shows the normalized complete-response magnitude (dB), including 200 ms averaging, at five operating points (A). Panel B compares predicted and empirical variance with an identity line; C plots their ratio against expected count (shape: probe conductance; colour: count).

Step 4: complete feature images

Pending plot. Pending Step 4. Planned file: `feature_images.png`.
Planned caption. Feature images at representative rates. Matched originals, library samples, and direct simulations are compared by pixel and image moments, zero fractions, and image agreement.

Step 5: mixed-rate decoder training

Pending plot. Pending Step 5. Planned file: `decoder_training.svg`.
Planned caption. Mixed-rate training. Training and validation loss and accuracy are shown by epoch and seed for both decoders; a rate histogram checks uniform sampling. Held-out accuracy is excluded.

Step 6: ANN psychometric curve and thresholds

Pending plot. Pending Step 6. Planned file: `ann_psychometric.svg`.
Planned caption. Decoder-relative psychometrics at 200 ms, spanning 0.25–25 Hz with a 0.25–5 Hz inset. Panel A compares frozen nonlinear and linear decoders at 1.2 µS, with chance and both thresholds; B compares nonlinear accuracy at 0.6, 1.2, and 2.4 µS. Uncertainty covers encoding and library draws and ANN seeds.

Step 7: handoff
Pending output. `decision.json` will report the decoder-relative edge, practical training floor, uncertainty, probe-conductance sensitivity, and the recommended rate range for a separate variable-rate PING training experiment.

Relation to prior work

Filtered conductance shot noise can violate fixed-time-constant models [**1**], and Gaussian approximations can miss skewed responses [**2**], supporting the empirical response library. Decoder performance is not absolute neural information [**3**], and linear and nonlinear decoders can differ [**4**], motivating a decoder-relative edge and both decoder diagnostics.

References

1. Wolff & Lindner: *Mean, Variance, and Autocorrelation of Subthreshold Potential Fluctuations Driven by Filtered Conductance Shot Noise.* Neural Computation, 2010. doi:10.1162/neco.2009.02-09-958
2. Brigham & Destexhe: *Nonstationary Filtered Shot-Noise Processes and Applications to Neuronal Membranes.* Physical Review E, 2015. doi:10.1103/PhysRevE.91.062102
3. Quian Quiroga & Panzeri: *Extracting Information from Neuronal Populations: Information Theory and Decoding Approaches.* Nature Reviews Neuroscience, 2009. doi:10.1038/nrn2578
4. Warland, Reinagel & Meister: *Decoding Visual Information From a Population of Retinal Ganglion Cells.* Journal of Neurophysiology, 1997. doi:10.1152/jn.1997.78.5.2336

Appendix A: linear-filter derivation

This appendix derives Equations 11–18 from the same AMPA synapse and subthreshold membrane used by the empirical probe. It replaces the finite Bernoulli drive with a continuous-time Poisson point process and linearizes the membrane around the mean drive at each calibration condition.

A.1 Stationary operating point

Represent pixel i's idealized spike train and synaptic conductance by

$$s_{i(t)} = \sum_k \delta(t - t_k), \quad \frac{dg_{i(t)}}{dt} = -\frac{g_{i(t)}}{\tau_{\mathrm{AMPA}}} + w_{\mathrm{probe}} s_{i(t)}. \quad \text{(A1)}$$

Here s_i (t) is a sum of Dirac impulses; t_k is spike k's time; δ is the Dirac delta; g_i (t) is AMPA conductance; τ_{AMPA} is its decay time; w_{probe} is conductance added per spike; and t is time. Equation A1 is the continuous-time counterpart of the decay-then-add update in Equation 3.

For stationary Poisson rate λ, the mean conductance satisfies

$$0 = -\frac{|g|_\lambda}{\tau_{\mathrm{AMPA}}} + w_{\mathrm{probe}}\lambda, \quad |g|_\lambda = \lambda w_{\mathrm{probe}} \tau_{\mathrm{AMPA}}. \quad \text{(A2)}$$

Here λ is encoding rate times pixel-intensity, and bar(g)$_\lambda$ is stationary mean conductance. The zero on the left states that the mean no longer changes with time.

Setting the mean membrane derivative in Equation 5 to zero gives

$$0 = g_{\mathrm{L,E}}(E_L - |v|_\lambda) + |g|_\lambda(E_e - |v|_\lambda), \quad |v|_\lambda = \frac{g_{\mathrm{L,E}} E_L + |g|_\lambda E_e}{g_{\mathrm{L,E}} + |g|_\lambda}. \quad \text{(A3)}$$

Here bar(v)$_\lambda$ is stationary mean voltage; $g_{\mathrm{L,E}}$ is leak conductance; and E_L and E_e are leak and AMPA reversal potentials. Other symbols follow Equation A2.

A.2 Local synapse-plus-membrane response

Write each signal as its operating point plus a small fluctuation:

$$g_i = |g|_\lambda + \delta g_i, \quad v_i = |v|_\lambda + \delta v_i, \quad s_i = \lambda + \delta s_i. \quad \text{(A4)}$$

Here δg_i, δv_i, and δs_i are complete conductance, voltage, and input perturbation variables around the means defined in Equations A2 and A3.

To derive Equation A5, substitute the conductance and input decompositions from Equation A4 into the synapse equation, Equation A1:

$$\frac{d(|g|_\lambda + \delta g_i)}{dt} = -\frac{|g|_\lambda + \delta g_i}{\tau_{\mathrm{AMPA}}} + w_{\mathrm{probe}}(\lambda + \delta s_i).$$

The operating point is stationary, so its time derivative is zero. Expanding the right-hand side then gives

$$\frac{d\delta g_i}{dt} = \left(-\frac{|g|_\lambda}{\tau_{\mathrm{AMPA}}} + w_{\mathrm{probe}}\lambda\right) - \frac{\delta g_i}{\tau_{\mathrm{AMPA}}} + w_{\mathrm{probe}} \delta s_i.$$

The parenthesized stationary terms cancel by Equation A2. The remaining perturbation dynamics are

$$\frac{d\delta g_i}{dt} = -\frac{\delta g_i}{\tau_{\text{AMPA}}} + w_{\text{probe}}\delta s_i. \quad \text{(A5)}$$

To derive Equation A6, substitute the conductance and voltage decompositions from Equation A4 into the membrane equation, Equation 5:

$$C_E \frac{d(|v|_\lambda + \delta v_i)}{dt} = g_{\text{L,E}}(E_L - |v|_\lambda - \delta v_i) + (|g|_\lambda + \delta g_i)(E_e - |v|_\lambda - \delta v_i).$$

As above, the stationary mean has zero time derivative. Expanding and grouping the right-hand side gives

$$C_E \frac{d\delta v_i}{dt} = \big(g_{\text{L,E}}(E_L - |v|_\lambda) + |g|_\lambda(E_e - |v|_\lambda)\big) - \big(g_{\text{L,E}} + |g|_\lambda\big)\delta v_i + (E_e - |v|_\lambda)\delta g_i - \delta g_i \delta v_i.$$

The first parenthesized group is zero by Equation A3. The final product is second order in the small perturbations and is omitted by the local linear approximation. This leaves

$$C_E \frac{d\delta v_i}{dt} = -\big(g_{\text{L,E}} + |g|_\lambda\big)\delta v_i + (E_e - |v|_\lambda)\delta g_i. \quad \text{(A6)}$$

Here C_E is excitatory-cell capacitance. All other symbols follow Equations A1–A4.

To derive Equation A7, take the Fourier transform of Equation A5. The derivative becomes $i\omega$ times the transformed signal:

$$i\omega \delta g_{i(\omega)} = -\frac{\delta g_{i(\omega)}}{\tau_{\text{AMPA}}} + w_{\text{probe}}\delta s_{i(\omega)}.$$

Move both conductance terms to the left, then divide by their common factor:

$$\left(i\omega + \frac{1}{\tau_{\text{AMPA}}}\right)\delta g_{i(\omega)} = w_{\text{probe}}\delta s_{i(\omega)},$$

$$\delta g_{i(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}}\delta s_{i(\omega)}. \quad \text{(A7)}$$

Equation A8 follows in the same way from Equation A6. Fourier transformation first gives

$$i\omega C_E \delta v_{i(\omega)} = -\big(g_{\text{L,E}} + |g|_\lambda\big)\delta v_{i(\omega)} + (E_e - |v|_\lambda)\delta g_{i(\omega)}.$$

Collect the voltage terms on the left:

$$\big(i\omega C_E + g_{\text{L,E}} + |g|_\lambda\big)\delta v_{i(\omega)} = (E_e - |v|_\lambda)\delta g_{i(\omega)}.$$

Dividing by the coefficient of $\delta v_i\ (\omega)$ gives

$$\delta v_{i(\omega)} = \frac{E_e - |v|_\lambda}{i\omega C_E + g_{\text{L,E}} + |g|_\lambda}\delta g_{i(\omega)}. \quad \text{(A8)}$$

Here ω is angular frequency, i is the imaginary unit, and each argument ω denotes a Fourier-domain signal.

Equations A5–A8 form a **linear time-invariant (LTI)** system after linearization about the operating point λ. Because the system has been linearized about the operating point, it is locally linear and time-invariant, so the standard transfer-function formalism applies. The canonical frequency-domain input–output relationship is

$$\delta v_{i(\omega)} = G_{\lambda(\omega)} \delta s_{i(\omega)}.$$

Here $G_\lambda(\omega)$ is the local transfer function relating input spike-train perturbations to membrane-voltage perturbations. For non-zero input, the input–output relationship can be rearranged as

$$G_{\lambda(\omega)} = \frac{\delta v_{i(\omega)}}{\delta s_{i(\omega)}}.$$

Thus the ratio follows from the standard input–output relationship rather than serving as an independent definition. To derive $G_\lambda(\omega)$, start from Equation A7:

$$\delta g_{i(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \delta s_{i(\omega)}.$$

Substitute this expression for $\delta g_i(\omega)$ into Equation A8:

$$\delta v_{i(\omega)} = \frac{E_e - |v|_\lambda}{i\omega C_E + g_{\text{L,E}} + |g|_\lambda} \left(\frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \delta s_{i(\omega)} \right).$$

Reorder the scalar factors and factor out $\delta s_i(\omega)$:

$$\delta v_{i(\omega)} = \left(\frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \cdot \frac{E_e - |v|_\lambda}{i\omega C_E + g_{\text{L,E}} + |g|_\lambda} \right) \delta s_{i(\omega)}.$$

Comparing this result with the canonical LTI relationship identifies the coefficient multiplying $\delta s_i(\omega)$ as the transfer function:

$$G_{\lambda(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \cdot \frac{E_e - |v|_\lambda}{i\omega C_E + g_{\text{L,E}} + |g|_\lambda}. \quad \text{(A9)}$$

Equation A9 is therefore the synaptic filter multiplied by the membrane filter, not an arbitrarily chosen ratio. Its λ-dependence captures the change in membrane gain and effective time constant with mean conductance.

A.3 Finite-window averaging

Start from the baseline-subtracted Step 1 feature in Equation 7:

$$z_i = \frac{1}{T} \int_0^T \left(v_{i(t)} - E_L \right) \mathrm{d}t.$$

Within the stationary linearized model used in this appendix, substitute the voltage decomposition from Equation A4:

$$z_i = \frac{1}{T} \int_0^T \left(|v|_\lambda + \delta v_{i(t)} - E_L \right) \mathrm{d}t.$$

The operating-point voltage and leak reversal potential are constant in time, so their integral is their difference multiplied by T. Dividing by T gives

$$z_i = |v|_\lambda - E_L + \frac{1}{T}\int_0^T \delta v_{i(t)}\,\mathrm{d}t.$$

Define the operating-point feature and its perturbation by

$$|z|_\lambda = |v|_\lambda - E_L, \quad \delta z_i = z_i - |z|_\lambda.$$

Subtracting the operating-point feature from the expanded expression for z_i cancels the constant term and leaves

$$\delta z_i = \frac{1}{T}\int_0^T \delta v_{i(t)}\,\mathrm{d}t. \quad \text{(A10)}$$

Thus δz_i is the fluctuation around the operating-point feature, obtained by averaging the voltage perturbation over the 200 ms presentation duration T. To derive the frequency response, represent the averaging operation as a causal rectangular kernel a_T (u), where

$$a_{T(u)} = \frac{1}{T}, \quad 0 \le u \le T,$$

and a_T (u) is zero outside that interval. Applying this kernel as a moving-average filter gives

$$\delta z_{i(t)} = \int_{-\infty}^{\infty} a_{T(u)}\delta v_{i(t-u)}\,\mathrm{d}u = \frac{1}{T}\int_0^T \delta v_{i(t-u)}\,\mathrm{d}u.$$

At the end of the presentation, set t equal to T. To put the resulting integral into the same form as Equation A10, introduce the new variable

$$t' = T - u, \quad \mathrm{d}t' = -\,\mathrm{d}u.$$

When u is zero, t' is T; when u is T, t' is zero. The substitution therefore reverses the integration limits. Its minus sign reverses them back:

$$\delta z_{i(T)} = \frac{1}{T}\int_0^T \delta v_{i(T-u)}\,\mathrm{d}u = -\frac{1}{T}\int_T^0 \delta v_{i(t')}\,\mathrm{d}t' = \frac{1}{T}\int_0^T \delta v_{i(t')}\,\mathrm{d}t'.$$

The name of an integration variable has no effect on the value, so replacing t' with t recovers Equation A10 exactly.

For any time-domain function f(u), use the Fourier-transform convention

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(u)\exp(-i\omega u)\,\mathrm{d}u.$$

Here hat(f) (ω) is the Fourier transform of f(u). Applying this definition with f(u) equal to the rectangular kernel a_T (u), and naming the result A_T (ω), gives

$$A_{T(\omega)} = \int_{-\infty}^{\infty} a_{T(u)}\exp(-i\omega u)\,\mathrm{d}u = \frac{1}{T}\int_0^T \exp(-i\omega u)\,\mathrm{d}u.$$

Evaluating the integral and rearranging gives

$$A_{T(\omega)} = \frac{\exp(-i\omega T) - 1}{-i\omega T} = \frac{1 - \exp(-i\omega T)}{i\omega T}.$$

At zero frequency, the continuous limit of $A_T(\omega)$ is one, as expected: averaging does not change a constant signal. By the convolution theorem, the averaging filter multiplies the voltage spectrum:

$$\delta z_{i(\omega)} = A_{T(\omega)} \delta v_{i(\omega)}.$$

Substitute the input–voltage relation from Equation A9:

$$\delta z_{i(\omega)} = A_{T(\omega)} G_{\lambda(\omega)} \delta s_{i(\omega)}.$$

Define the complete local input–feature response by the canonical LTI relationship

$$\delta z_{i(\omega)} = H_{\lambda(\omega)} \delta s_{i(\omega)}.$$

Comparing the coefficients multiplying $\delta s_i(\omega)$ identifies $H_\lambda(\omega)$. The two components of Equation A11 are therefore

$$A_{T(\omega)} = \frac{1-\exp(-i\omega T)}{i\omega T}, \quad H_{\lambda(\omega)} = A_{T(\omega)} G_{\lambda(\omega)}. \quad \text{(A11)}$$

Here $A_T(\omega)$ is the averaging-window response, exp is the exponential function, and $H_\lambda(\omega)$ is the complete response from input spike-train perturbation to averaged-voltage perturbation.

For the Bode comparison, the zero-frequency-normalized magnitude of the complete input–feature response is

$$B_{H(\omega)} = 20 \log_{10}\left(\frac{\left|H_{\lambda(\omega)}\right|}{\left|H_{\lambda(0)}\right|}\right). \quad \text{(A12)}$$

Here B_H is magnitude in decibels; $|\cdot|$ is complex magnitude; and $\log_{10}$ is the base-ten logarithm. Only H_λ is plotted because it includes the averaging window and is the response used to predict the ANN feature variance.

A.4 Poisson-noise variance

A centred, unit-amplitude Poisson point process has a flat two-sided input power spectral density:

$$S_{\mathrm{in}(\omega)} = \lambda. \quad \text{(A13)}$$

Here $S_{\mathrm{in}}(\omega)$ is input power per unit angular frequency and λ is mean spike rate. Passing this noise through Equation A11 gives

$$S_{z(\omega)} = \left|H_{\lambda(\omega)}\right|^2 S_{\mathrm{in}(\omega)}. \quad \text{(A14)}$$

Here $S_z(\omega)$ is predicted feature-noise power and $|H_\lambda(\omega)|^2$ is the transmitted fraction of input-noise power. Under the two-sided angular-frequency convention, total predicted variance is

$$\mathrm{Var}_{\mathrm{linear}(z)} = \frac{1}{2\pi}\int_{-\infty}^{\infty} \left|H_{\lambda(\omega)}\right|^2 S_{\mathrm{in}(\omega)}\, \mathrm{d}\omega. \quad \text{(A15)}$$

Here $\mathrm{Var}_{\mathrm{linear}}(z)$ is predicted feature variance and π is the circle constant; the remaining symbols follow Equations A11–A14. This result is exact for the stated linear stationary model, but only approximate for the finite, discrete, conductance-dependent probe. Step 3 therefore compares it with empirical variance rather than assuming agreement.